

PROXY PATTERN

Image.java:

```
public interface Image {  
    void display();  
}
```

ProxyImage.java:

```
public class ProxyImage implements Image {  
    private String filename;  
    private ReallImage realimg;  
    public ProxyImage(String filename) {  
        this.filename=filename;  
    }  
    @Override  
    public void display() {  
        if(realimg==null) {  
            realimg=new ReallImage(filename);  
        }  
        realimg.display();  
    }  
}
```

ReallImage.java:

```
public class ReallImage implements Image {  
    private String filename;  
    public ReallImage(String filename) {  
        this.filename=filename;  
        loadImage();  
    }  
    private void loadImage() {  
        System.out.println("Loading "+filename+" from remote server...");  
    }  
    @Override  
    public void display() {  
        System.out.println("Displaying "+filename);  
    }  
}
```

```
}  
}
```

ProxyPatternTest.java:

```
public class ProxyPatternTest {  
    public static void main(String[] args) {  
        Image img=new ProxyImage("nature.jpg");  
        System.out.println("First Display:");  
        img.display();  
        System.out.println();  
        System.out.println("Second Display:");  
        img.display();  
    }  
}
```

